

Education

University of California, San Diego | San Diego, CA

SEP 2024 - JUN 2026

M.S. in Computer Science

GPA: 4.00

Relevant Coursework: Deep Learning, Machine Learning, NLP, ML Systems, Artificial Intelligence, Recommender Systems, OS

University of California, Berkeley | Berkeley, CA

AUG 2020 - MAY 2024

B.A. in Computer Science & B.A. in Applied Math

GPA: 3.90

Professional Experience

Machine Learning Engineer Intern | Qualcomm

JUN 2025 -

Teaching Assistant | CSE 153/253: Machine Learning for Music @ UCSD

MAR 2025 - JUN 2025

Teaching Assistant | CSE 234: Data Systems for ML @ UCSD

DEC 2024 - MAR 2025

• Construct 3 brand-new programming assignments for students mat-mul optimization, backpropagation using PyTorch and Triton

• Host office hours, grade projects and exams, maintain Ruby & Jekyll course website, assign relevant papers to students for reading

Research Intern | Hao AI Lab @ UCSD

APR 2024 -

• Develop cutting-edge LLM monitoring system, training SAEs for new sequence-level early exit strategies in Python & PyTorch

• Configure new CLLM models to be compatible with qLoRA fine-tuning on Jacobi trajectories, decreasing memory usage over 3x

Machine Learning Engineer Intern | Accretional

JUN 2024 - SEP 2024

• Design and implement RAG vector embedding model to ground LLM with respect to cloud infrastructure

• Create command-line tool in Golang for indexing and vectorization of files, semantic & lexical search of keywords

• Fully construct VSCode application in Svelte and TypeScript to help users easily implement and deploy APIs in less than 5 seconds

Software Engineer Intern | Hinge Health

MAY 2023 - AUG 2023

• Overhauled and integrated a tedious (100+ hour) manual user data reset workflow process

• Designed 5+ endpoints and event consumers in TypeScript and Ruby on Rails, restructuring and building new GraphQL queries

• Presented solutions for over 10 UI/UX features in React, linked repos with RabbitMQ event emitting, deployed in Kubernetes

Full Stack Intern | Cisco Meraki

MAY 2022 - AUG 2022

• Spearheaded a full stack project to create uplink ping stat graphs, helping 500,000 customers visualize wireless trends over time

• Collected ping stats data from over 4 million wireless internet nodes, using Scala scripts to build a grabber and Jenkins to test

• Conducted experiments with global LT & SQL queries across 600 shards to mitigate the size of rapidly-growing tables

Software Engineer Intern, Student Tutor | Breakout Mentors

JUN 2021 - AUG 2021

Projects & Research

MusicRFM, interpretability of music using recursive feature machines

JAN 2025 -

CoT Reasoning with Sparse Autoencoders

JAN 2025

Learner, a full stack Flask / React app that prompts & fine-tunes LLMs to help with learning languages

SEP 2024 - JAN 2025

Convolutional Neural Net, AlexNet, UNet implementation

MAR 2024 - APR 2024

• Engineered CNN, AlexNet, UNet and variants, including construction of layers, activations, autodiff in NumPy, PyTorch

Insite, a React, NestJS & TypeScript app that connects high schoolers to professionals

JUN 2023 - SEP 2023

• Utilized AWS S3 buckets for file storage, assembled 40+ REST API endpoints for tasks like search, connections, and profile pages

• Built custom Docker containers and PSQl databases, integrating them with CRA frontend & backend repos with networks

Regulus, a Golang secure file sharing system

MAR 2023 - MAY 2023

• Employed over 5 cryptographic tools and libraries like HashKDF, PBKDF, UUID, HMACs for authentication & access revocation

NumC, a C++ version of numpy matmul, optimized 40x using loop unrolling, SimD, OpenMP, caches

OCT 2021 - DEC 2021

Skills

Python, SQL, JavaScript, TypeScript, React, PyTorch, TensorFlow, Java, Golang, Kubernetes, AWS, Ruby on Rails, Scala, Docker, C, C#, C++, RabbitMQ, GraphQL, HTML, CSS, Figma, Jenkins, PHP, Unity, Git / GitLab / GitHub, Unix, Jira, JupyterLab